MATH 345 Section 1.1: Vectors

Vectors and notation

Scalar: real number.

Vector: ordered list of scalars; $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$.

Row / column notation: $\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ or $\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$.

Zero vector: $\mathbf{0} = (0, \dots, 0)$.

Equality: same dimension and equal corresponding entries: $\mathbf{v} = \mathbf{w} \iff v_i = w_i$ for every i.

Displacement: from $P(p_1, \dots, p_n)$ to $Q(q_1, \dots, q_n)$, $\overrightarrow{PQ} = Q - P = (q_1 - p_1, \dots, q_n - p_n)$.

Vector operations

For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and $t \in \mathbb{R}$:

Addition: $\mathbf{v} + \mathbf{w} = (v_1 + w_1, \dots, v_n + w_n)$.

Scalar multiplication: $t\mathbf{v} = (tv_1, \dots, tv_n)$.

Linear combination: $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$, with $\mathbf{v}_i \in \mathbb{R}^n$; scalars c_i are the **coefficients**.

Convex combination: linear combination with $c_i \geq 0$ for every i and $c_1 + \cdots + c_k = 1$.

Norm and dot product

Length / Euclidean norm: $\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}$.

Dot product: $\mathbf{v} \cdot \mathbf{w} = v_1w_1 + \cdots + v_nw_n$ (a scalar).

Identity: $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$.

Euclidean distance: $\operatorname{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

Angle and orthogonality

Angle theorem: for nonzero $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, the angle $\theta \in [0, \pi]$ satisfies

$$\cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|}.$$

Orthogonal: $\mathbf{v} \cdot \mathbf{w} = 0$; for nonzero vectors, equivalent to $\theta = 90^\circ$.

Cosine similarity: $\operatorname{cosim}(\mathbf{v}, \mathbf{w}) = \dfrac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|} = \cos\theta \in [-1, 1]$.

Domain restriction: angle and cosine similarity undefined if either vector is $\mathbf{0}$.